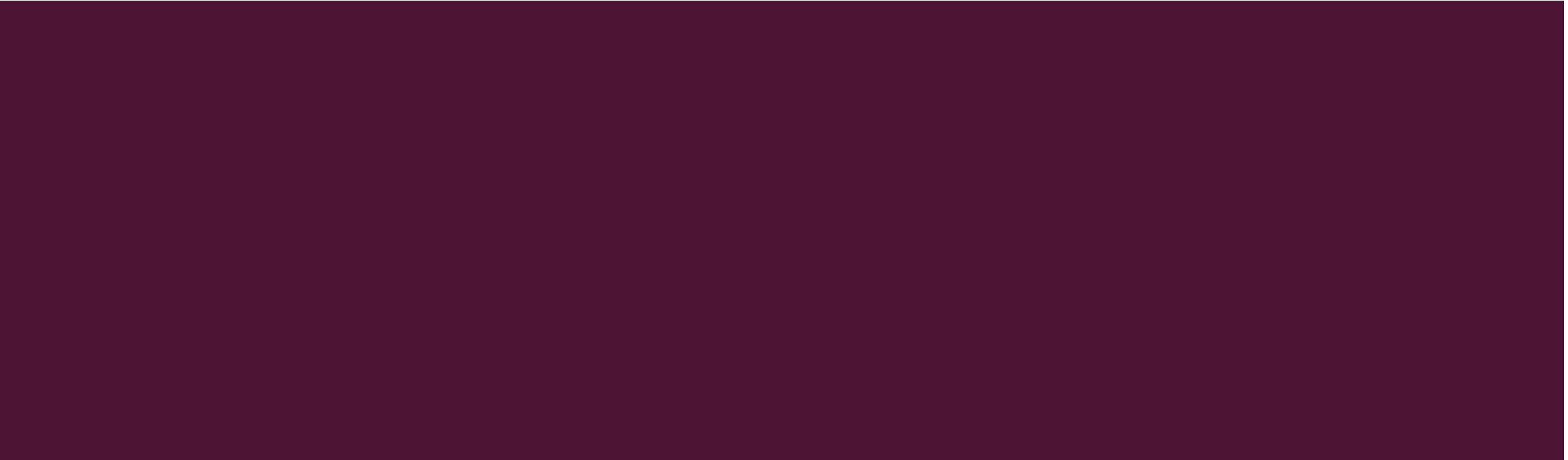

WHO WROTE THIS TWEET? DETECTING HUMAN VS. LLM-GENERATED TEXT ON TWEETFAKE

DENIS MALASI

DEEP LEARNING 24/25



MOTIVATION & PROBLEM

- LLM text is everywhere on social media
- Trust, moderation, and spam are at risk
- Need to tell **human vs AI** reliably
- Tweets are short, noisy → weak signals



PROJECT GOALS

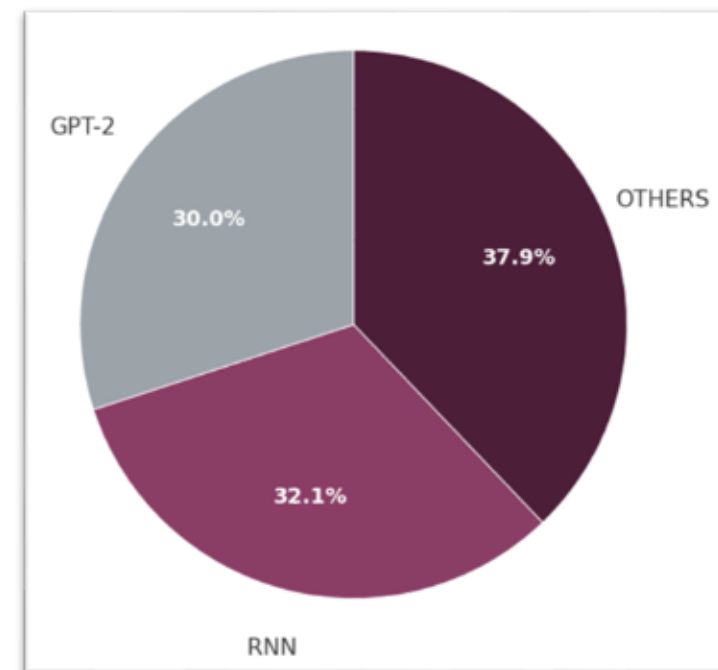
- Use the **TweepFake** English tweet dataset
- Train & compare **SVM, CNN, Transformer**
- Main metrics: **Accuracy & F1-score**
- **Interpretability:** top discriminative words



DATASET

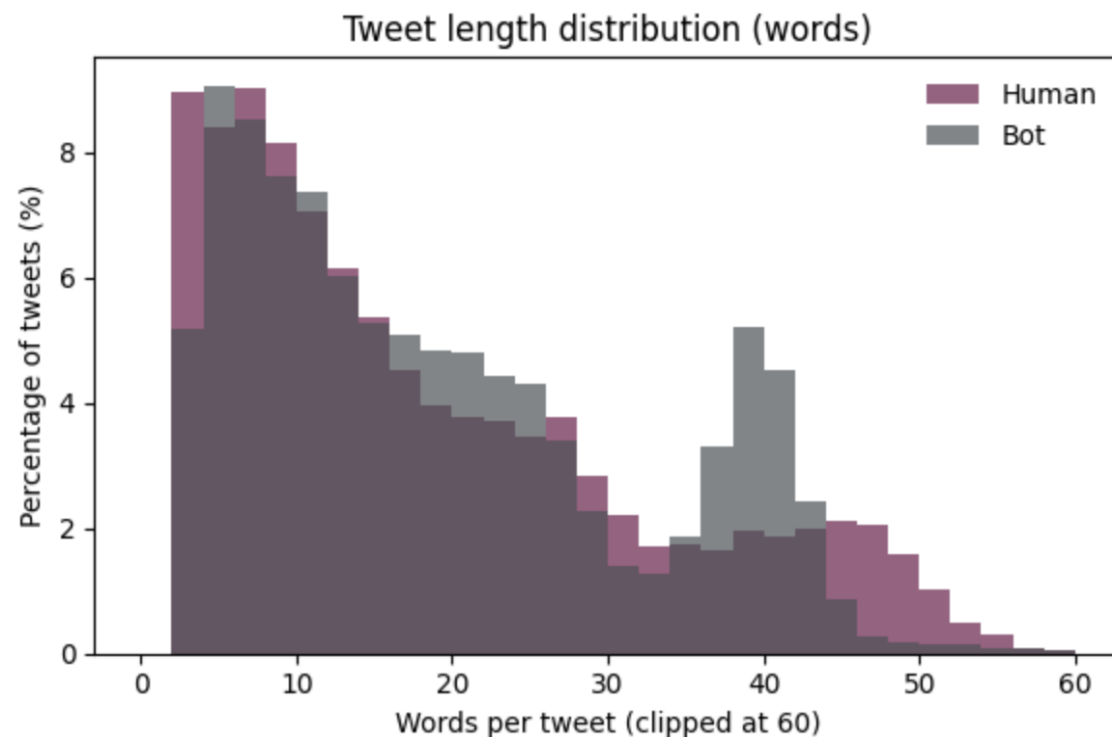
TWEEPFake: HOW IT'S BUILT

- **17 real Twitter users** (English) selected as targets
- Their tweets scraped → **human class**
- **LLM / text models** (GPT-2, RNN, Markov, LSTM...) trained on those tweets to mimic the style
- **23 bot accounts** posted the generated tweets



TWEEPFake: STATS & SPLITS

- **25,572 tweets** → Perfectly balanced
- Avg words: **18.4 (H) vs 18.8 (B)**
- Avg chars: **117.5 (H) vs 104.5 (B)**
- 81 / 9 / 10 (Train/Val/ Test)
 - Balanced in each set



STRUCTURAL DIFFERENCES (HUMAN VS BOT)

Feature	Human	Bot	Human/Bot ratio
Links	0.43 / tweet	0.03 / tweet	17.5 × more
Mentions	0.49 / tweet	0.13 / tweet	3.8 × more
Hashtags	0.08 / tweet	0.045 / tweet	1.8× more
Emojis	0.31 / tweet	0.18 / tweet	1.7× more
Punctuation Marks	6.2 / tweet	3.8 / tweet	1.6× more
Uppercase Words	0.43 / tweet	0.32 / tweet	1.3× more
Lexical Diversity	0.95	0.92	+3%



SVM



TF-IDF & LINEAR SVM

- **TF-IDF**
 - **TF** = term frequency in the current tweet
 - **IDF** = rarity across *all* tweets
 - **TF-IDF** = $TF \times IDF \rightarrow$ distinctive terms $\uparrow$, ubiquitous terms $\downarrow$
- **Linear SVM**
 - Margin-based **linear** separator on sparse TF-IDF vectors
 - Hinge loss + C regularization $\rightarrow$ large margin, good generalization

TEXT CLEANING & NORMALIZATION

- Lowercase + HTML unescape + Compress multiple spaces
- Replace URLs → __url__, mentions → __user_mention__
- Remove “#” but keep the hashtag word

Raw: "OMG!!! Check this out: <https://t.co/xYz> @John #AI #DeepLearning"

Clean: "omg !!! check this out __url__ __user_mention__ ai deeplearning"

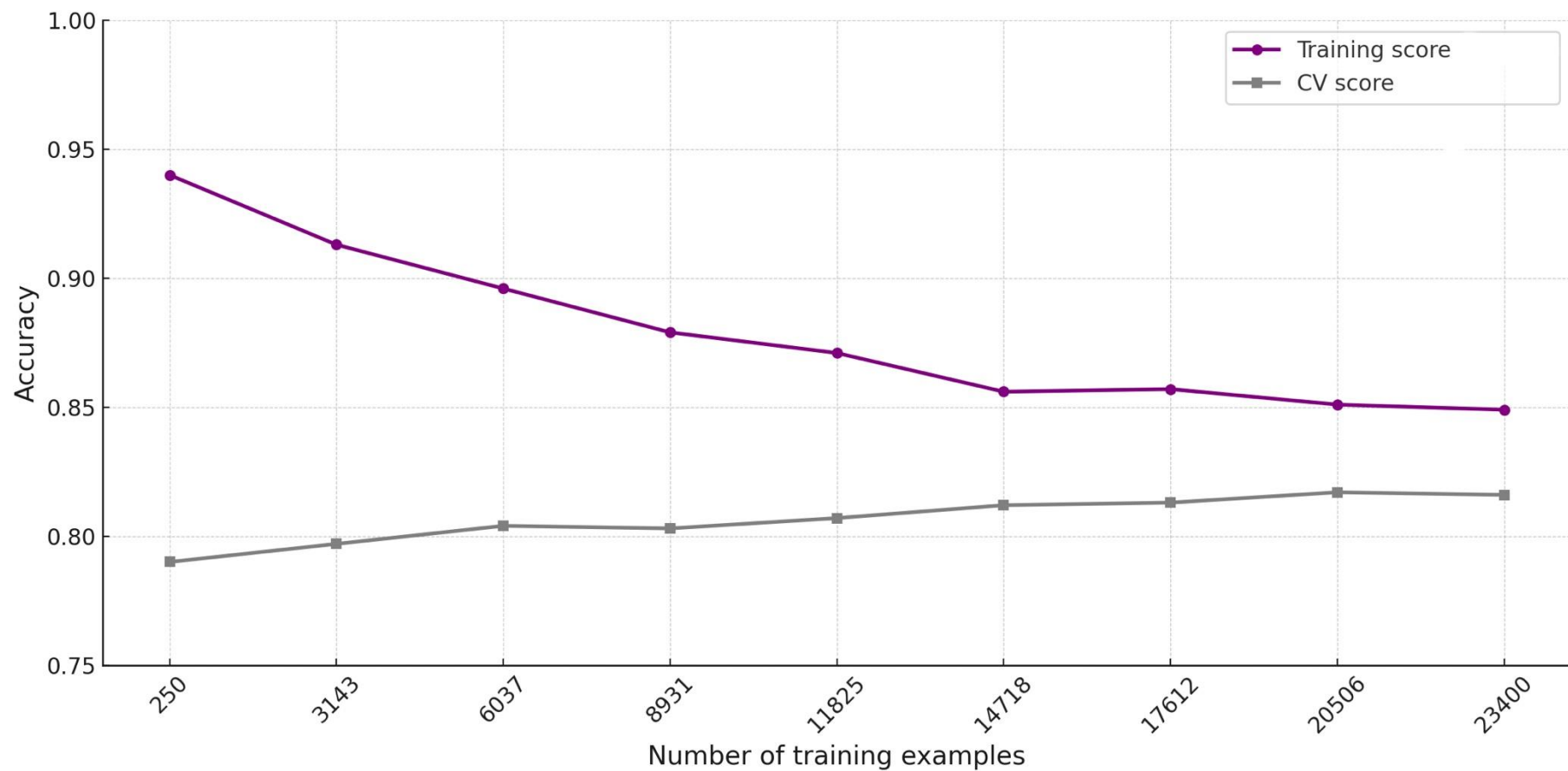
HYPERPARAMETER TUNING (5-FOLD CV)

- 5-fold stratified CV on Train+Val:
 - TF-IDF: **(1,3)** n-grams with **25.000 features**
 - Linear SVM: **C=1**, **hinge** loss and no stopwords removal

- Mean CV accuracy: **82.5%**

N-Gram Range	CV accuracy
(1,1)	81.3%
(1,2)	81.5%
(1,3)	82.5%
(1,4)	82.4%

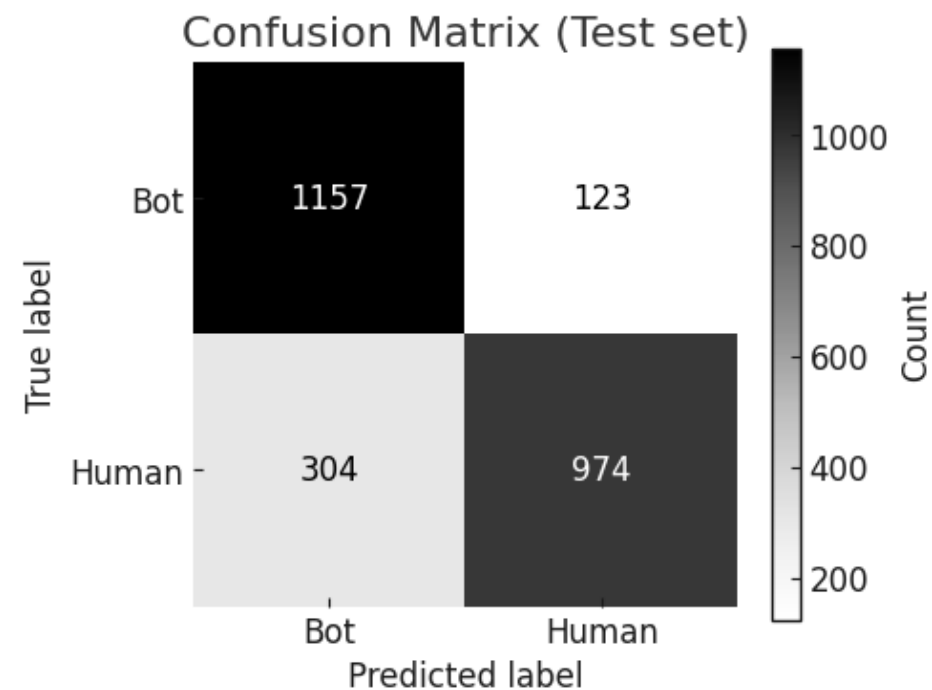
SVM LEARNING CURVE: TRAINING VS. CV SCORE



FINAL TEST PERFORMANCE

Metric	Bot	Human
Precision	79.2%	88.8%
Recall	90.4%	76.2%
F1-score	84.4%	82.0%

- Final Accuracy on test set: 83.3%
- High bot recall (90.4%) → **few missed bots**
- Human precision high (88.8%) → **few false alarms**



TOP 15 PREDICTIVE N-GRAMS FOR BOT VS. HUMAN TWEETS

N-gram (Bot)	Coefficient
the __url__	+2.029
the people	+1.806
__src_tag__	+1.752
to __user_mention__	+1.669
rt __user_mention__	+1.591
install	+1.582
files	+1.568
horrible	+1.564
java	+1.503
obama	+1.501
config	+1.456
believe me	+1.446
ridiculous	+1.432
care about	+1.418
disaster	+1.399

N-gram (Human)	Coefficient
__url__	-2.681
iran	-2.431
covid	-2.325
annoying	-2.182
👍	-2.056
discovered	-1.878
covid 19	-1.872
followers	-1.836
about the wrong	-1.777
wiping	-1.753
__user_mention__ bro	-1.748
goku	-1.621
lunch	-1.609
drunk	-1.599
basketball	-1.588

EXAMPLE TWEETS & PREDICTION OUTCOMES

- **True Human:**

- 'hahah 😄👍__url__' → score: -2.578

- **True Bot**

- 'I thank the government in the people of the development is the programme of the republic of mangal for the state' → score: +2.852

- **False Positive** (human predicted as bot)

- 'I am seeking 250k investment in exchange for 10 equity' → score: +0,882

- **False Negative** (bot predicted as human)

- 'old pinned tweet__url__' → score: -1.558

SVM CLASSIFIER – LIMITATIONS

- **Uneven performance by bot type:**
 - GPT-2: 76.0%
 - RNN: 94.7%
 - Others: 95.1%
- **Ambiguous or overly formal/promotional tweets** often misclassified
- **Generic phrasing** reduces discrimination in edge cases

The image shows the CNN logo, which consists of the letters "CNN" in a dark red, serif font. The logo is positioned to the right of a vertical dark red line. The entire logo is centered between two horizontal gray bars that span the width of the image.

CNN

CHAR-CNN OVERVIEW

- **Input:** raw tweet as a sequence of characters
- **Embedding:** character embedding layer
- **Convolutions:** multiple 1D filters to capture local character patterns
- **Activation:** tanh applied after each convolution
- **Pooling:** global max-over-time to extract the strongest feature per filter
- **Dropout:** regularization to prevent overfitting
- **Classification:** bot vs. human prediction

PREPROCESSING & HYPERPARAMETER TUNING

- **Text cleaning:**
 - Lowercase, HTML unescape, compress multiple spaces
- **Parameters explored:**
 - Embedding dimension, max length, convolution kernel sizes, number of filters, dropout probability, learning rate, batch size
- **Tuning strategy:**
 - Grid search with **5-fold stratified cross-validation** on train+val
 - Early stopping (patience = 5 epochs) within each fold

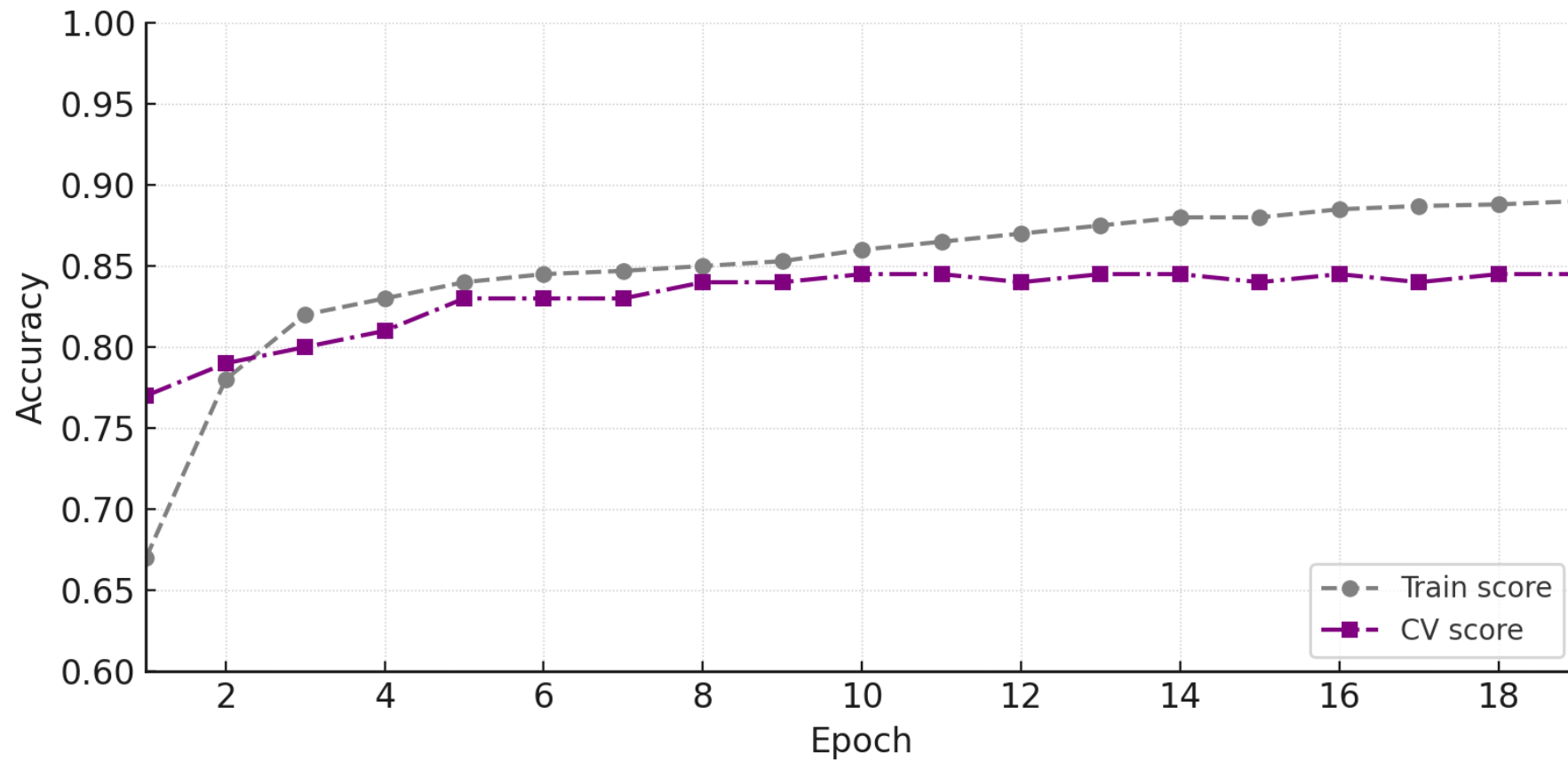
CHAR-CNN FINAL SETUP

Parameter	Value
Embedding dimension	64
Max lenght	280
Convolution kernel size	[3,4]
Number of filters	128
Dropout	0.6
Learning rate	5×10^{-4}
Batch size	128
Optimizer	Adam
Loss	CrossEntropy

- Mean 5-Fold CV Accuracy: 84.24 %

Kernel	Accuracy
[3]	83.6%
[3, 4]	84.2%
[3, 4, 5]	84.0%
[4, 5, 6]	84.1%
[1, 2, 3]	83.9%

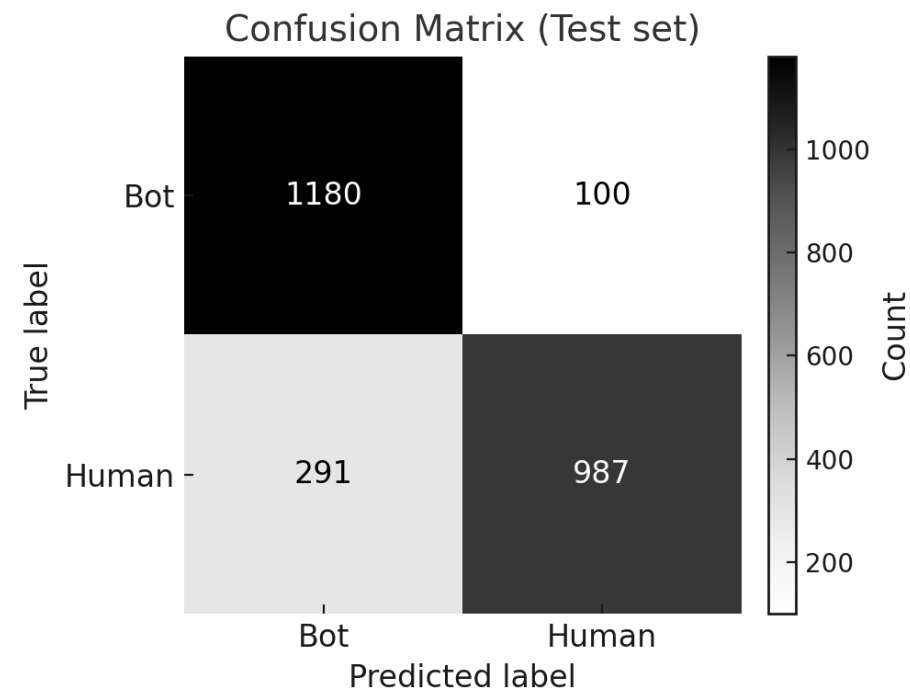
LEARNING CURVES: TRAINING VS. CV SCORE



FINAL TEST PERFORMANCE

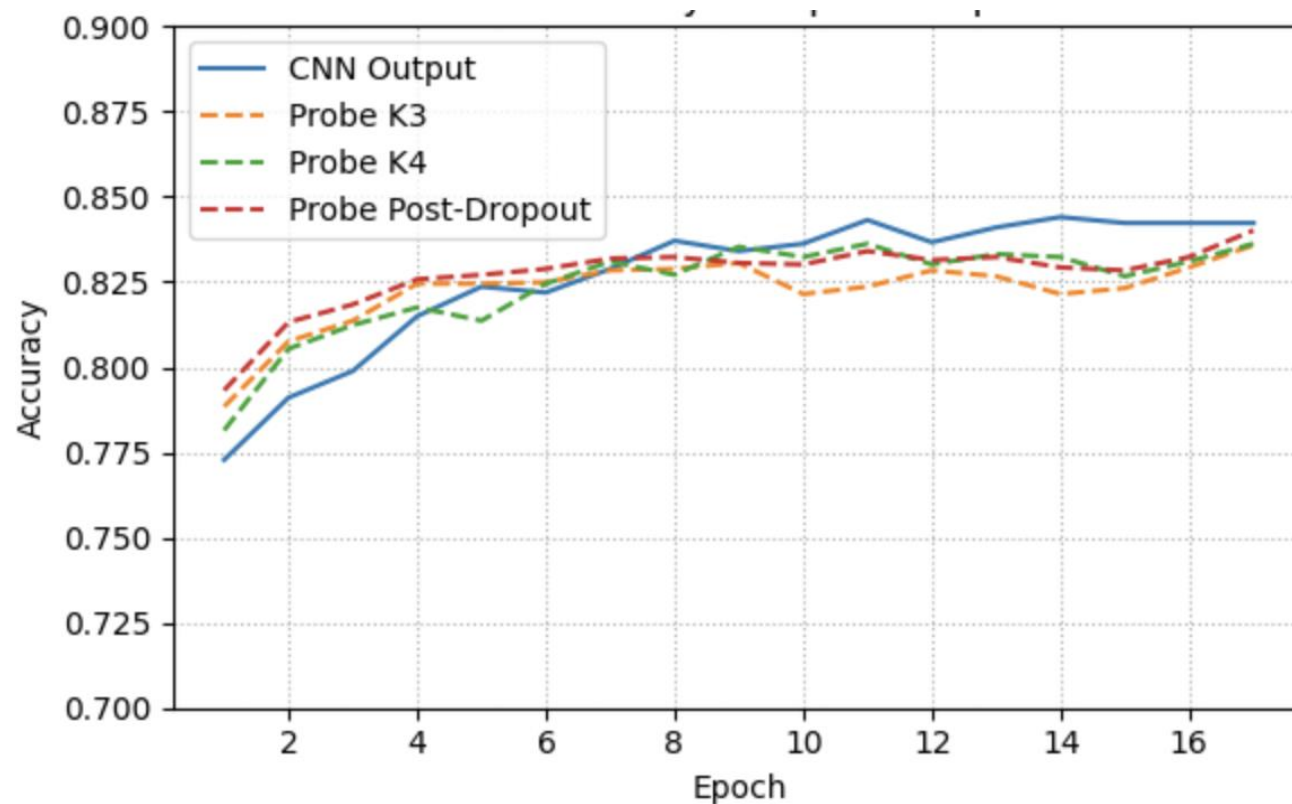
Metric	Bot	Human
Precision	80.2%	90.8%
Recall	92.2%	77.2%
F1-score	85.8%	83.4%

- Final Accuracy on test set: **84.7%**
- High bot recall (92.2%) → **few missed bots**
- Human precision high (90.8%) → **few false alarms**



INTERMEDIATE REPRESENTATION PROBES

- **Three probes placed at:**
 - **After Conv1** (kernel=3)
 - **After Conv2** (kernel=4)
 - **Post-Dropout**
- Even first conv layers are discriminative
- Post-dropout nearly matches full model



TOP 15 PREDICTIVE WORDS FOR BOT VS. HUMAN TWEETS

N-gram (Bot)	Coefficient
better	+15.0
have	+14.1
programme	+13.5
https://t.co/j3hsvmm4u4	+13.4
respects	+13.4
https://t.co/hocyb8nxnu	+13.1
getting	+12.4
committed	+12.0
refactoring	+11.8
luther	+10.8
been	+10.8
great	+10.6
what	+10.0
west	+10.0
unfollow	+9.9

N-gram (Human)	Coefficient
popular	-4.0
people	-3.9
wonderful	-3.0
yee	-2.3
tweets	-2.2
picked	-2.1
simplicity	-2.1
orphans	-2.0
explan	-2.0
pissed	-1.8
tomorrow	-1.8
understood	-1.8
socially	-1.8
ignored	-1.6
sis	-1.6

CNN – LIMITATIONS & IMPROVEMENTS OVER SVM

- **Bot-type performance (Test):**
 - GPT-2: **82.3%** (vs. 76.0% SVM) → significant improvement
 - RNN: 96.4% (vs. 94.7% SVM)
 - Others: 95.7% (vs. 95.1% SVM)
- **Improvements over SVM** → better handling of fluent, GPT-2 text
- **Limitations:**
 - **Lower human accuracy:** humans still misclassified more often than bots ($\approx 77\%$ recall)
 - **Higher compute cost** compared to lightweight SVM + TF-IDF



TRANSFORMER-BASED CLASSIFIER (BERT)

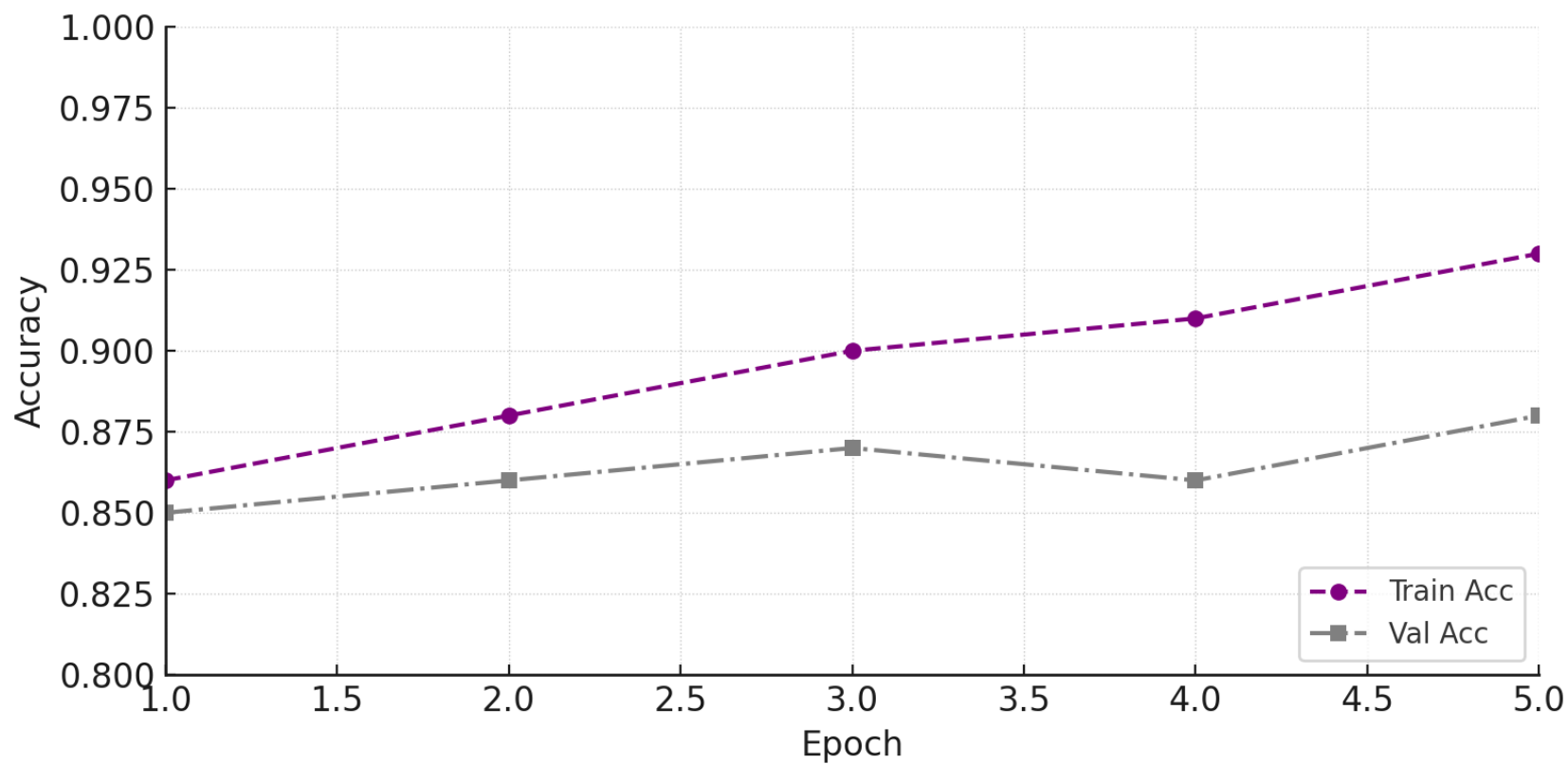
TRANSFORMER-BASED CLASSIFIER (BERT)

- **Model:** BERT-base-uncased (12 encoder layers, 768-dim hidden)
- **Input:** WordPiece tokens with special *[CLS]* and *[SEP]* markers
- **Core layers:** 12 Transformer encoders using self-attention to capture context across the tweet
- **Representation:** use the final *[CLS]* token embedding as a summary of the entire tweet
- **Classification head:** dropout + linear layer → bot vs. human prediction

FINE-TUNING SETUP & HYPERPARAMETERS

- **Input:** max 128 WordPiece tokens ([CLS] + text + [SEP])
- **Batch size:** 16
- **Epochs:** 5
- **Learning rate:** 2×10^{-5} (AdamW)
- **Weight decay:** 0.05
- **Dropout (classification head):** 0.6
- **Layer freezing:**
 - Frozen: all BERT parameters except encoder layers 10–11, pooler, and classifier head

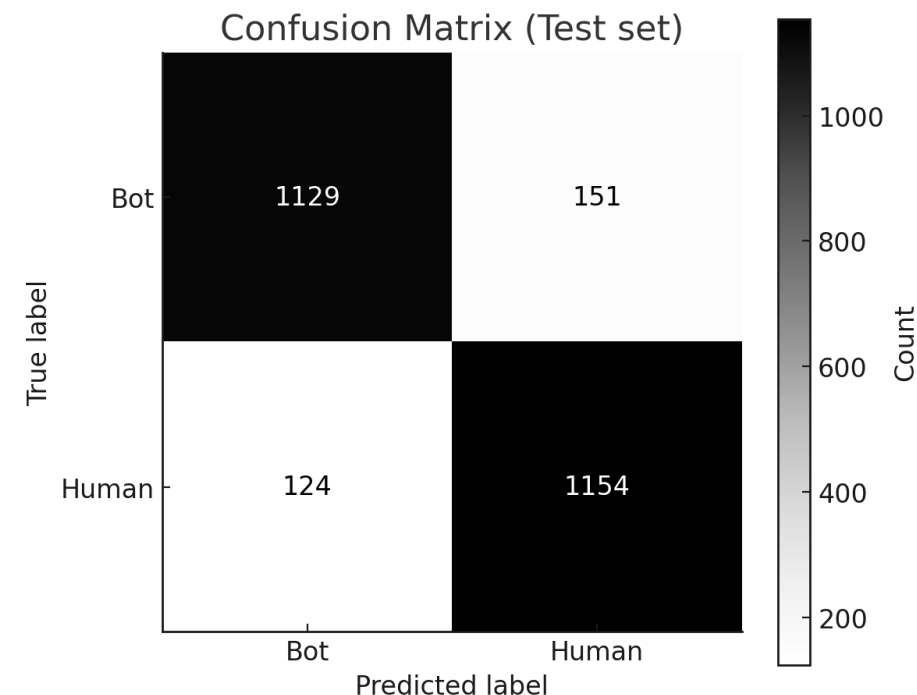
LEARNING CURVES: TRAINING VS. VALIDATION ACCURACY



FINAL TEST PERFORMANCE

Metric	Bot	Human
Precision	90.1%	88.4%
Recall	88.2%	90.3%
F1-score	89.1%	89.4%

- Final Accuracy on test set: **89.2%**
- Bot precision jumps nearly 10 points → **fewer false alarms**
- Human recall improves over 13 points → **fewer missed humans**



BERT – IMPROVEMENTS OVER CHAR-CNN & REMAINING LIMITATIONS

- Improvements over Char-CNN:
 - Overall accuracy + **4.5 %** (84.7% → 89.2%)
 - Better on GPT-2 bots: from 82.3% → **83.5%**
 - Deep contextual understanding
 - Minimal feature engineering
- Remaining Limitations:
 - Compute & memory cost
 - **Rare/unseen tokens:** WordPiece may split unknown slangs